UNIT - III : STATISTICAL QUALITY CONTROL AND TOTAL

I. Quality and Quality Control

* QUALITY = fitness for use (Juran); conformance to requirements (Crosby); the totality of features and characteristics of a product or service that bear on its ability to satisfy stated or implied needs (ISO). Dimensions (Garvin): performance, features, reliability, conformance, durability, serviceability, aesthetics, perceived quality.
* QUALITY CONTROL (QC) = the operational techniques and activities used to fulfil requirements for quality - inspection, testing and corrective action to ensure that products meet specifications at least cost.
* QUALITY ASSURANCE (QA) = planned and systematic activities (the system - procedures, audits, documentation) implemented to GIVE CONFIDENCE that the quality requirements will be met. QC is product oriented (detects defects); QA is process / system oriented (prevents defects).

Quality Control	Quality Assurance
Detection - correct defects	Prevention - avoid defects
Product oriented	Process / system oriented
Line function (inspection, testing)	Staff function (audits, standards)
Reactive	Proactive

Importance of quality assurance

* Customer satisfaction and loyalty; reputation and competitiveness; lower cost (less scrap, rework, warranty claims); higher productivity; legal and safety compliance; access to markets (ISO certification);

reduced liability; employee morale.

* Cost of quality: PREVENTION cost (training, planning), APPRAISAL cost (inspection, testing), INTERNAL failure (scrap, rework), EXTERNAL failure (returns, warranty, lost goodwill). Spend more on prevention - total cost falls.

Inspection

* Checking the product against specification. 100% (complete) inspection - for critical parts, small lots, costly and boring, fatigue errors, destructive tests impossible. SAMPLING inspection - for large lots, destructive testing, cheaper, faster. Types: incoming, in-process, final; first piece, floor, centralised.

2. Statistical Quality Control (SQC)

* SQC = application of STATISTICAL methods (probability theory, sampling) to control and improve quality; started by W. A. SHEWHART (Bell Labs, 1924, control chart); sampling plans by Dodge and Romig. Two branches: (1) Process control - CONTROL CHARTS, (2) Product control - ACCEPTANCE SAMPLING.

Causes of variation

* No two items are exactly alike. CHANCE (random / common) causes: many small unavoidable causes, always present, predictable pattern - process is 'in statistical control'. ASSIGNABLE (special) causes: definite, identifiable (worn tool, bad material, operator error), cause big variation - must be found and removed.

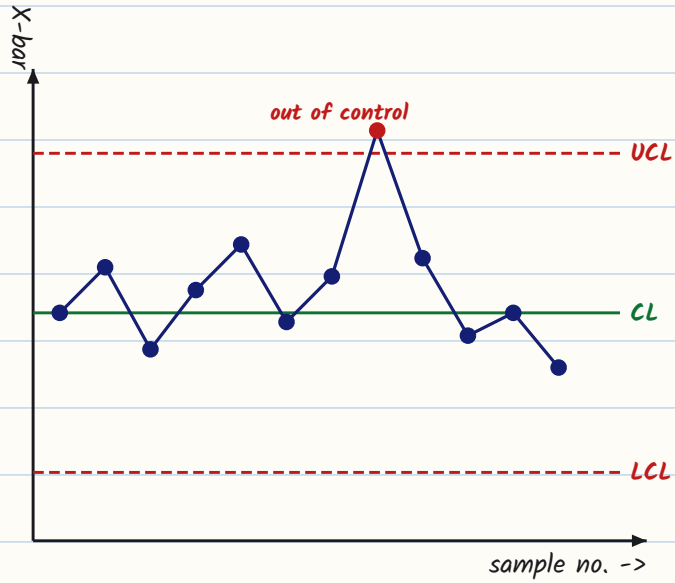
Statistical basis

* Quality characteristic follows NORMAL distribution: mean mu, standard deviation sigma. Within mu +- 3 sigma lies 99.73% of items (mu +- 1

sigma 68.27%, +- 2 sigma 95.45%). Hence control limits are set at +- 3 sigma ('three sigma limits'): the chance of a point outside by chance alone is only 0.27%.

* Variables = measurable characteristics (length, weight, temperature) -> X-bar, R, s charts. Attributes = go / no-go characteristics (defective or not, number of defects) -> p, np, c, u charts.

3. Control Charts for Variables



* Control chart = graph of sample statistic against sample number with a CENTRE LINE (CL), UPPER and LOWER control limits (UCL, LCL). Points inside limits with random pattern -> process in control. Point outside limits, run of 7 on one side, trend, cycles -> out of control: search for assignable cause.
* Procedure: take k samples (usually 20 - 25) each of n items (n = 4 or 5); compute X-bar and R (range = max - min) for every sample; compute grand mean X-double-bar and average range R-bar; compute limits; plot; analyse.

X-bar and R charts

$$\text{X-double-bar} = \text{sum}(\text{X-bar}_i) / k \text{ , R-bar} = \text{sum}(R_i) / k$$

$$\text{X-bar chart: UCL} = \text{X-dbar} + A_2\ \text{R-bar} \text{ , LCL} = \text{X-dbar} - A_2\ \text{R-bar}$$

$$\text{R chart: UCL} = D_4\ \text{R-bar} \text{ , LCL} = D_3\ \text{R-bar}$$

n	A2	D3	D4	d2	A3	B3	B4
2	1.880	0	3.267	1.128	2.659	0	3.267
3	1.023	0	2.574	1.693	1.954	0	2.568
4	0.729	0	2.282	2.059	1.628	0	2.266
5	0.577	0	2.114	2.326	1.427	0	2.089
6	0.483	0	2.004	2.534	1.287	0.030	1.970
7	0.419	0.076	1.924	2.704	1.182	0.118	1.882

* Estimate of process standard deviation sigma = R-bar / d2. The R chart is examined first (variability), then the X-bar chart (central tendency).

X-bar and s charts

$$\text{X-bar chart: UCL} = \text{X-dbar} + A_3\ \text{s-bar} \text{ , LCL} = \text{X-dbar} - A_3\ \text{s-bar}$$

$$\text{s chart: UCL} = B_4\ \text{s-bar} \text{ , LCL} = B_3\ \text{s-bar}$$

* s (sample standard deviation) is used instead of R when sample size is LARGE (n > 10) - range loses efficiency; computation is heavier, but s is a better measure of spread.

Numerical example (X-bar and R)

10 samples of size n = 5 gave sum of sample means = 500.0 and sum of

ranges = 30.0 (A2 = 0.577, D3 = 0, D4 = 2.114).

$$\text{X-dbar} = 500/10 = 50.0 \; ; \; \text{R-bar} = 30/10 = 3.0$$

$$UCL(X) = 50 + 0.577 \times 3 = 51.73 \; ; \; LCL(X) = 50 - 1.731 = 48.27$$

$$UCL(R) = 2.114 \times 3 = 6.342 \; ; \; LCL(R) = 0 \times 3 = 0$$

Process sigma = 3.0 / 2.326 = 1.29. Specification 50 +- 4: Cp = (USL - LSL)/(6 sigma) = 8/7.74 = 1.03 (just capable).

Applications

* Control of dimensions of machined parts (diameter, thickness), weight / fill of packages, tablet weight in pharmaceuticals, temperature or pH in chemical processes, electrical parameters of components, service times. Used to monitor, detect shifts early, decide when to adjust, establish process capability and reduce scrap.

4. Control Charts for Attributes

Chart	Plots	Sample size	Limits
p chart	Fraction defective p	Variable or constant	p-bar +- 3 sqrt(p-bar (1 - p-bar) / n)
np chart	Number defective	Constant	np-bar +- 3 sqrt(np-bar (1 - p-bar))
c chart	Number of defects per unit	Constant (one unit)	c-bar +- 3 sqrt(c-bar)
u chart	Defects per unit	Variable	u-bar +- 3 sqrt(u-bar / n)

Eg: 20 samples of 100 items had 60 defectives in all: p-bar = 60/2000 =

0.03; UCL = 0.03 + 3 sqrt(0.03 x 0.97 / 100) = 0.03 + 0.0512 = 0.0812; LCL = 0.03 - 0.0512 < 0 -> taken as 0.

c chart eg: average 9 defects per fabric roll: UCL = 9 + 3 x 3 = 18; LCL = 9 - 9 = 0.

5. Acceptance Sampling - Attribute Sampling Inspection

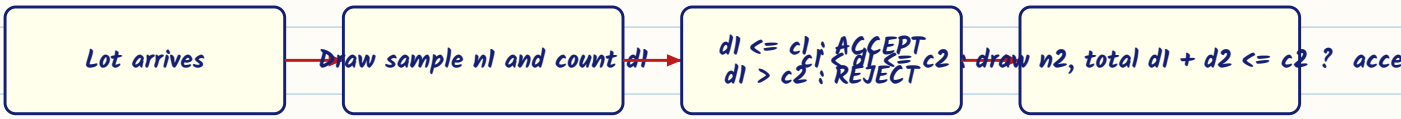
* Decision to ACCEPT or REJECT a lot is taken from the inspection of a random SAMPLE. Plans are defined by sample size n and acceptance number c (max defectives allowed in sample). Lot is accepted if defectives found <= c, otherwise rejected (or 100% inspected).

Single sampling plan

* Take ONE sample of size n from lot size N. Count defectives d. If d <= c -> accept lot; if d > c -> reject. Simple, easy to administer, but sample size (and inspection cost) is large and gives only one chance.

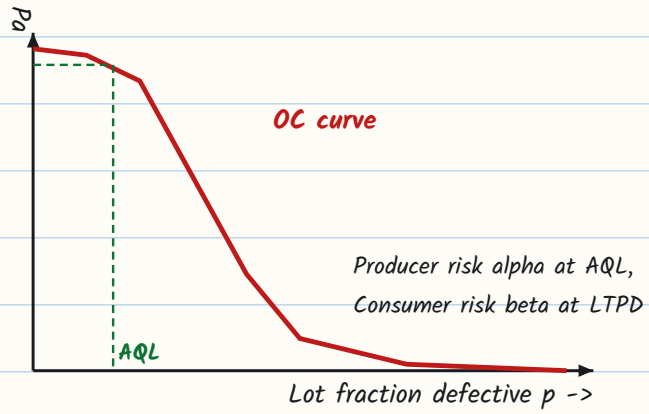
Double sampling plan

* Plan (n1, c1, n2, c2): take FIRST sample n1, with d1 defectives. If d1 <= c1 -> accept; if d1 > c2 -> reject; if c1 < d1 <= c2 -> take SECOND sample n2 with d2: if d1 + d2 <= c2 -> accept, else reject. Gives a second chance to the lot, average inspection is less than single sampling for very good / very bad lots, psychologically more acceptable, but is more complex.



Single sampling	Double sampling
One sample	Up to two samples
Larger sample, higher inspection cost	Lower average sample number (ASN)
Simple administration	More complex records
One decision chance	Second chance for borderline lots

* MULTIPLE (sequential) sampling: units tested one by one until decision. OC CURVE (Operating Characteristic): plot of probability of acceptance Pa against lot fraction defective p; shows plan's discriminating power. AQL = Acceptable Quality Level (good lot, producer risk alpha ~ 5%); LTPD / RQL = Lot Tolerance Percent Defective (bad lot, consumer risk beta ~ 10%); AOQ = average outgoing quality, AOQL = its maximum.



Pa (single plan, Poisson) = sum from d = 0 to c of e^{-np} $(np)^d$ / d!

Eg: n = 50, c = 1, p = 2%: np = 1.0; Pa = e^{-1} (1 + 1) = 0.7358 (73.6% chance of acceptance).

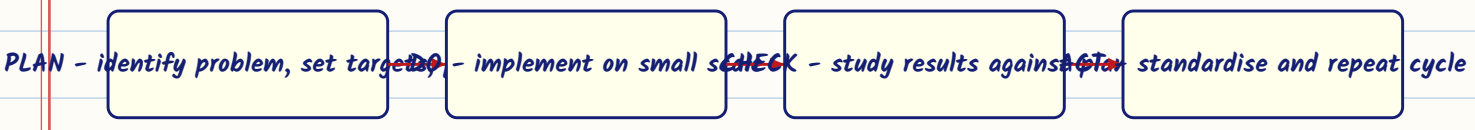
6. Total Quality Management (TQM)

* TQM = a management philosophy and system in which ALL members of the organisation continuously improve processes, products, services and culture to meet or exceed CUSTOMER expectations. 'Total' - everyone and every function; 'Quality' - meeting customer requirements; 'Management' - leadership and systems. Pioneers: W. Edwards Deming (14 points, PDCA cycle), Joseph Juran (trilogy), Philip Crosby (zero defects), Ishikawa (quality circles, fishbone), Feigenbaum (TQC), Taguchi (loss function).

Elements (principles) of TQM

* 1. CUSTOMER focus (internal and external). 2. LEADERSHIP and top management commitment. 3. EMPLOYEE involvement, empowerment and teamwork; training. 4. PROCESS approach. 5. CONTINUOUS IMPROVEMENT (Kaizen). 6. Fact based decisions - data, SPC. 7. SUPPLIER partnership. 8. Quality cost, benchmarking, quality function deployment (QFD), FMEA, JIT, 5S.

Continuous improvement - PDCA / Kaizen



* KAIZEN = small, continuous, incremental improvement by everybody (Japanese: 'change for the better'). Tools: the 7 QC tools - check sheet, histogram, Pareto diagram (80:20 rule), cause-and-effect (Ishikawa / fishbone) diagram, scatter diagram, control chart, stratification / flow chart. Also 5S (Seiri sort, Seiton set in order, Seiso shine, Seiketsu standardise, Shitsuke sustain).

Zero defect concept

* Philip B. Crosby (1960s, Martin Marietta): 'Quality is free'; performance standard = ZERO DEFECTS - do it right the FIRST time (cost of quality = cost of non-conformance). Four absolutes: quality = conformance to requirements; prevention, not appraisal; performance standard zero defects; measurement = price of nonconformance. Motivation by commitment, awareness, error cause removal and ZD Day. Related: Poka-Yoke (mistake proofing, Shigeo Shingo).

Quality circles

* A small group (6 - 12) of employees from the same work area who meet VOLUNTARILY and regularly (about 1 hour per week) to identify, analyse and solve work related quality problems and present solutions to management. Originated in Japan (Ishikawa, 1962, JUSE); in India, BHEL (1981). Structure: members, leader, facilitator, steering committee. Tools: brainstorming, Pareto, fishbone, data collection.

Benefits: employee involvement and motivation, problem solving, cost reduction, improved communication and teamwork.

Implementation of TQM and applications

* Steps: top management commitment and quality policy -> create quality council -> assess current status -> training of all employees -> select projects, form teams -> measure and benchmark -> implement improvements -> reward recognition -> review and sustain. Obstacles: lack of commitment, resistance to change, poor training, short-term thinking, no customer focus.
* Applications: manufacturing (automobiles - Toyota), services (banks, hospitals), education, IT, government, hotels. Benefits: higher customer satisfaction, lower defects and cost, market share, employee morale.

7. ISO Quality Systems

* ISO (International Organisation for Standardisation, Geneva, 1947) - ISO 9000 family: standards for QUALITY MANAGEMENT SYSTEMS (QMS). ISO 9000 - fundamentals and vocabulary; ISO 9001:2015 - requirements (the one certified); ISO 9004 - performance improvement guidelines; ISO 19011 - auditing.
* ISO 9001:2015 clauses (High Level Structure): 4 Context of the organisation, 5 Leadership, 6 Planning (risk based thinking), 7 Support, 8 Operation, 9 Performance evaluation (audit, review), 10 Improvement. Seven QM principles: customer focus, leadership, engagement of people, process approach, improvement, evidence-based decision making, relationship management.
* Other: ISO 14001 (Environmental Management), ISO 45001 (Occupational Health and Safety), ISO 27001 (Information Security),

ISO/TS 16949 -> IATF (automotive), ISO 13485 (medical devices), ISO 22000 (food safety). Certification by an accredited body after audit; surveillance audits each year; re-certification every 3 years. Benefits: documented procedures, international recognition, customer confidence, fewer errors, continual improvement.

8. Six Sigma

* SIX SIGMA = a data driven, disciplined methodology and philosophy for eliminating DEFECTS and VARIATION in any process, aiming at near-perfect quality: 3.4 DEFECTS PER MILLION OPPORTUNITIES (DPMO). Developed at MOTOROLA (Bill Smith, Mikel Harry, 1986); popularised by General Electric (Jack Welch).
* Meaning of sigma level: the process mean is 6 standard deviations from the nearest specification limit (with allowance for a 1.5 sigma long term shift of the mean).

Sigma level	DPMO	Yield %
1	691,462	30.9
2	308,538	69.1
3	66,807	93.3
4	6,210	99.38
5	233	99.977
6	3.4	99.99966

DPMO = (number of defects / (number of units x opportunities per unit)

Basic concepts

* DMAIC (improve existing process): DEFINE the problem and customer

needs (project charter, VOC); MEASURE current performance (data collection, baseline sigma); ANALYSE root causes (fishbone, Pareto, hypothesis tests, regression); IMPROVE (solutions, DOE, pilot); CONTROL (SPC, control plan, standardise). DMADV / DFSS (new process design): Define, Measure, Analyse, Design, Verify.

* Belts (roles): Champion (top executive sponsor), Master Black Belt (coach), BLACK BELT (full time project leader), GREEN BELT (part time, team member), Yellow / White Belt. Key terms: CTQ (Critical To Quality), VOC (Voice of Customer), process capability Cp / Cpk, SIPOC, defect, opportunity, yield, rolled throughput yield. LEAN SIX SIGMA = Lean (remove waste) + Six Sigma (reduce variation).

Cp = (USL - LSL) / (6 sigma) ; Cpk = min((USL - mean)/(3 sigma), (m

Revision: SQC = control charts (process) + acceptance sampling (product). X-bar limits = X-dbar +- A2 R-bar; R chart: D3, D4. TQM = customer focus + continuous improvement. Six Sigma = 3.4 DPMO, DMAIC. ISO 9001 = QMS.